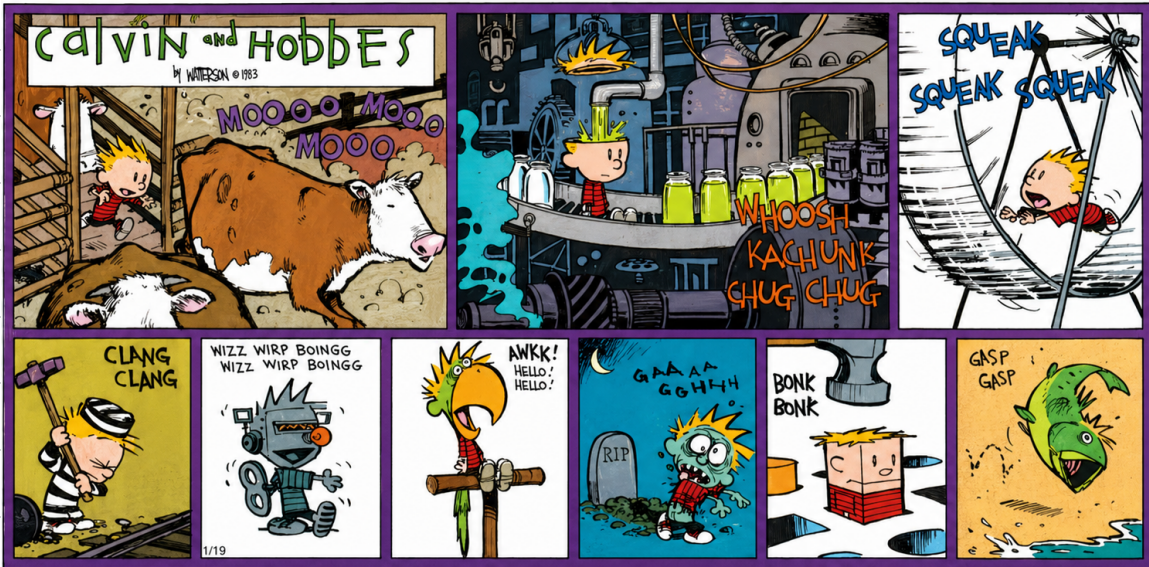
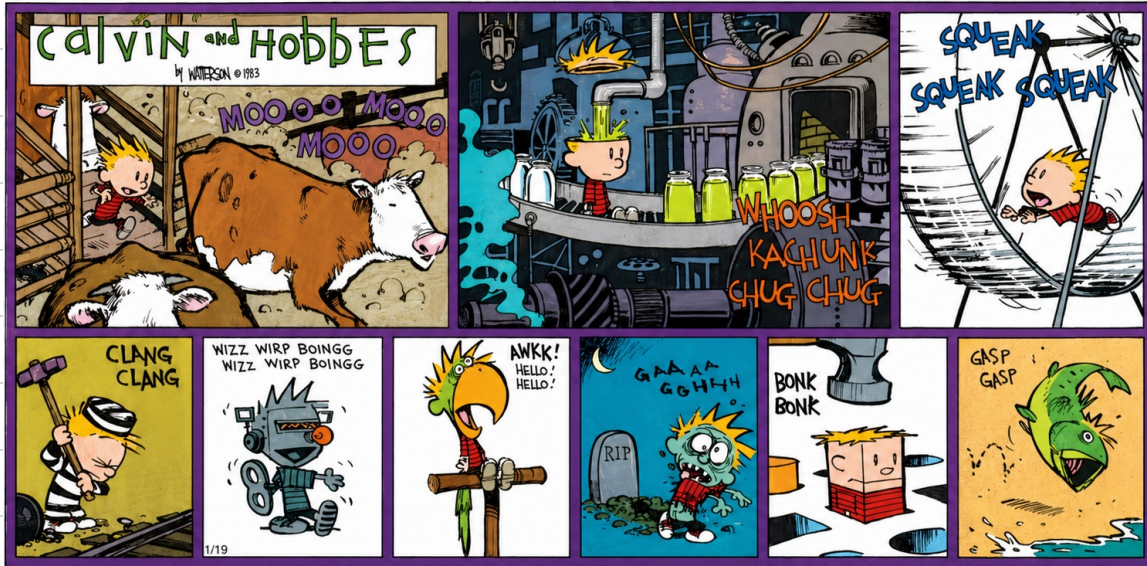
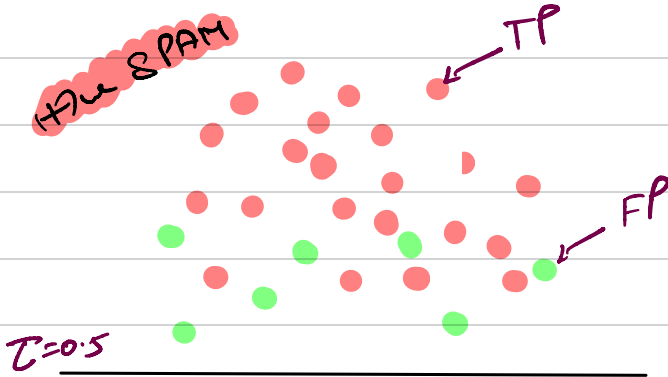




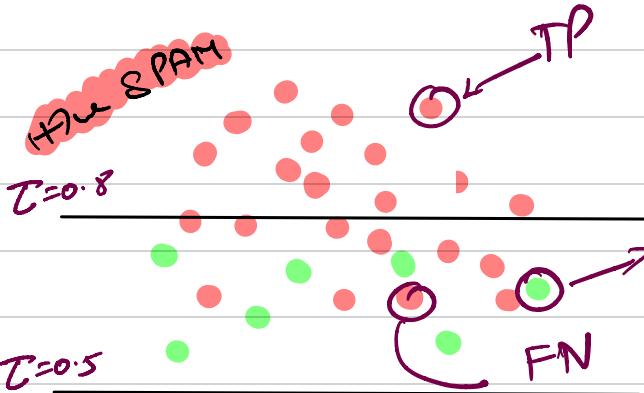
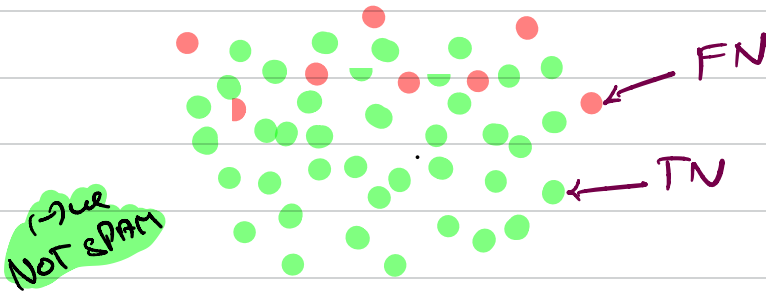


Precision Recall - Variability



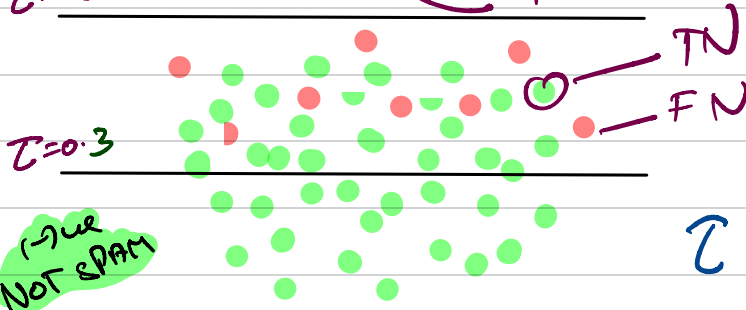
$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{FN + TP}$$



$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{FN + TP}$$



$$Z \propto \text{Precision} < \frac{1}{\text{Recall}}$$

$C_1 - 0.3$
 $C_2 - 0.2$
 $C_3 - 0.5 \rightarrow C-3 - \text{Predict's}$

F1-Score

	P	R	Avg.	Harmonic me.
π_1	0.8	0.3	0.55	0.43
π_2	0.2	0.9	0.55	0.33
π_3	0.7	0.4	0.55	0.51

$$\text{Harmonic Mean}(a, b) = \frac{2ab}{a+b} = \underline{\text{F1 Score}}$$

Pos - Fraud
 Neg - N. Fraud

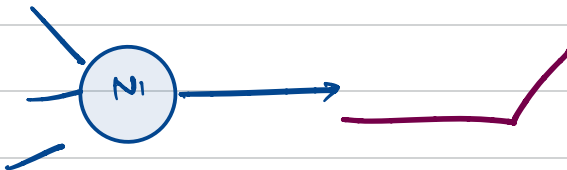
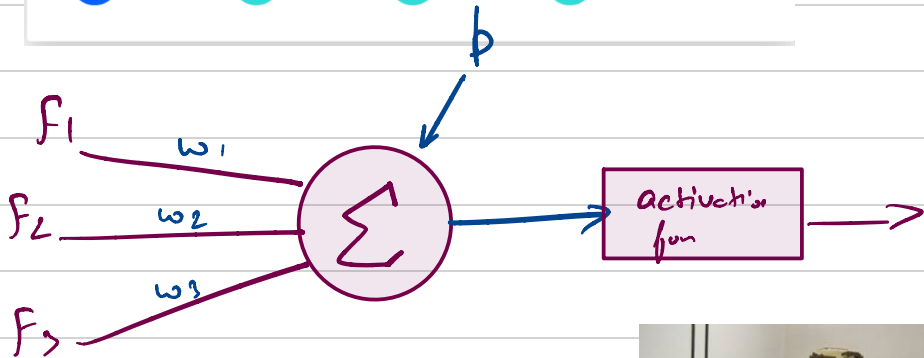
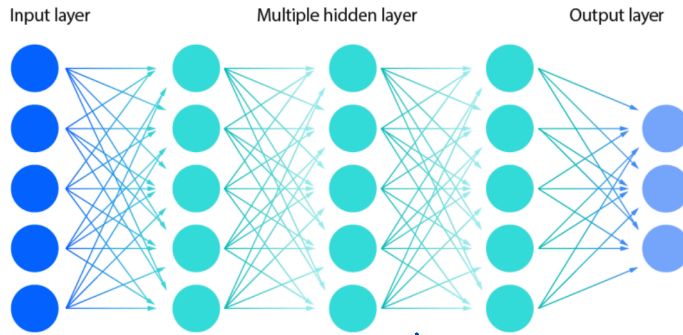
FP $\rightarrow \downarrow$

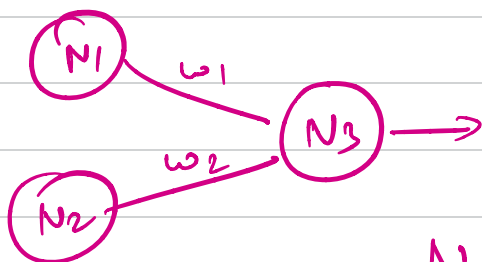


FN $\rightarrow \downarrow$

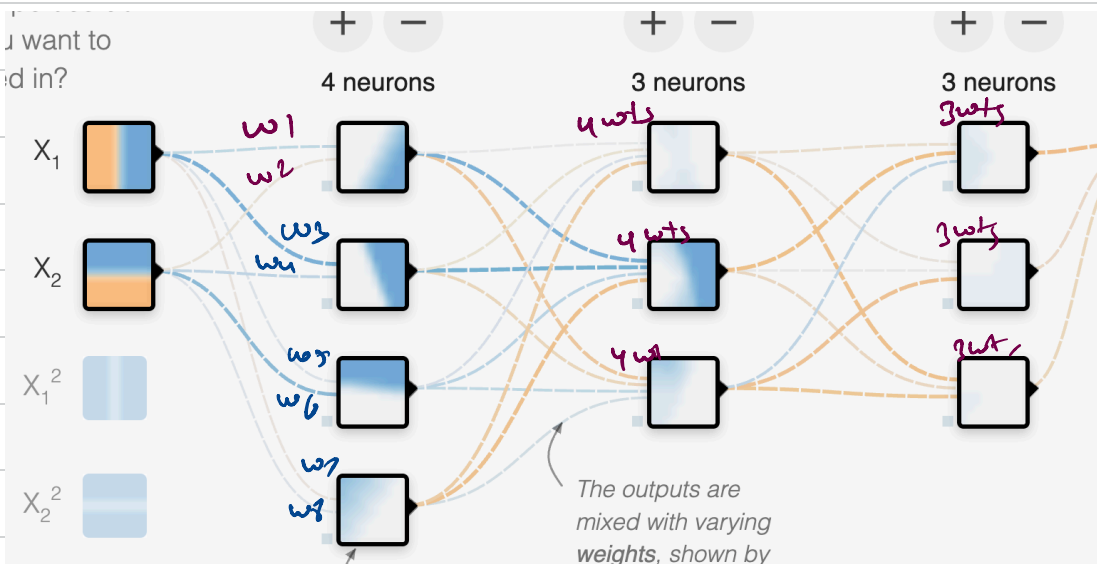
Neural Networks - Re

Deep neural network



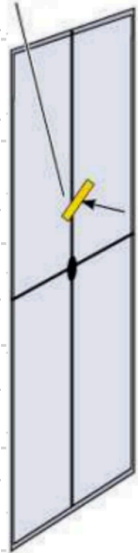


$$N_3 = \text{Relu}(w_1 \times N_1 + w_2 \times N_2)$$



Experimental setup

Light bar stimulus
projected on screen



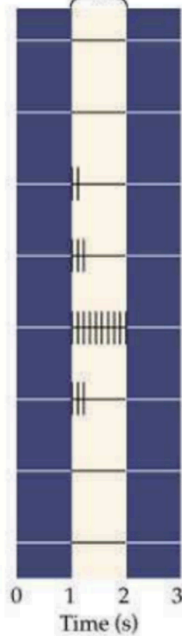
Recording from visual cortex

Record

Stimulus
orientation



Stimulus
presented



$$N = (\sum w_i x_i + b)$$

→ f_{in}

→ Not f

1 Billion samples



1000

1000

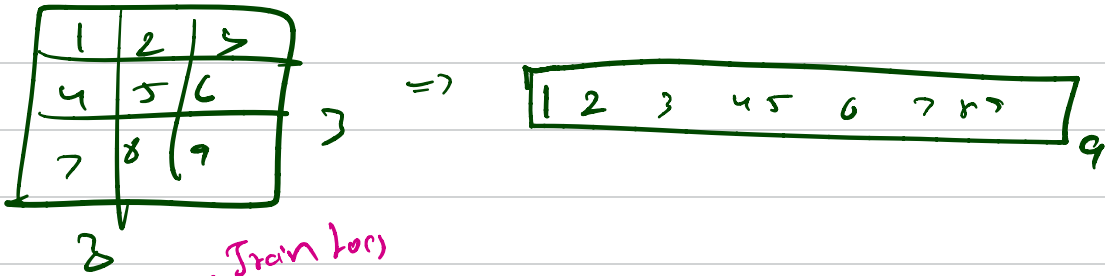


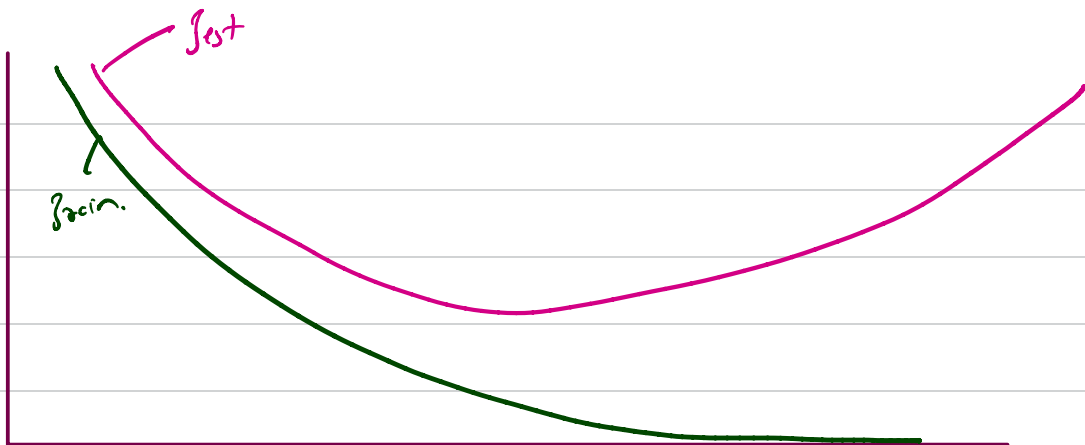
Epoch, batch size

100 Rows $\rightarrow$ batch of 10 rows

10 batch $\rightarrow$ 10 iteration

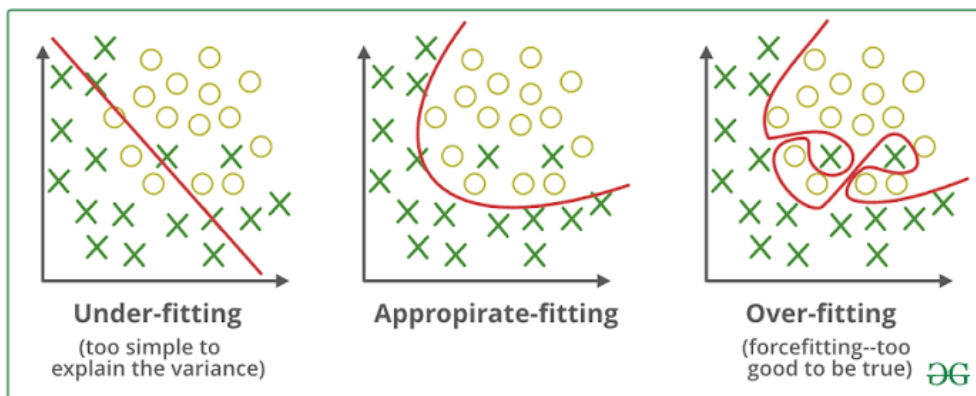
Epoch $\rightarrow$ 100 Rows
once $\rightarrow$ 10 batches / iteration.





1

1024	→	L_1
512	—	L_2
256	—	L_3
128	—	L_4



different:

Loss function

vs

Metric

Post model
is
trained

Gradients

→

$\frac{\partial L}{\partial \omega}$

have a
mean

Not diff-

$$MAPE = \frac{1}{N} \sum_{i=1}^n$$

$$\left| \frac{y_i - \hat{y}}{y_i} \right|$$

% change
in y_i

100 Rows $\rightarrow$ batches - 20 row

w - randomly

$$y = w_1 f_1 + w_2 f_2 + w_0$$

$\square B_1 \rightarrow \text{model} \rightarrow \hat{y} \rightarrow \text{gradient-update}$

$\square B_2 \rightarrow \text{model}_{w_1} \rightarrow \hat{y} \rightarrow \text{gradient-update}$

$\square B_3 \rightarrow \text{model}_{w_2} \rightarrow \hat{y} \rightarrow \text{gradient-update}$

$\square B_4 \rightarrow \text{model}_{w_2} \rightarrow \hat{y} \rightarrow \text{gradient-update}$

$\square B_5 \rightarrow \text{model}_{w_2} \rightarrow \hat{y} \rightarrow \text{gradient-update}$

$$\hat{y} = w_0 + w_1 m_1 + w_2 m_2 + \dots + w_n m_n$$

$$\hat{f}(m) = w^T x + w_0$$

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2$$

$$\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2}, \dots, \frac{\partial L}{\partial w_n}, \frac{\partial L}{\partial w_0}$$

$$L = \frac{1}{n} \sum_{i=1}^m \left(y_i - (\omega_1 x_1 + \omega_2 x_2 + \dots + \omega_n x_n + \omega_0) \right)^2$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^m \left(y_i - \omega_1 x_1 - \omega_2 x_2 - \dots - \omega_n x_n - \omega_0 \right)^2$$

$$\frac{\partial L}{\partial \omega_1} \Rightarrow \frac{1}{n} \times \sum_{i=1}^m 2 \times \left(y_i - \omega_1 x_1 - \omega_2 x_2 - \dots - \omega_n x_n - \omega_0 \right) \times (-x_1)$$

$$= \frac{2}{n} \sum_{i=1}^m (y_i - \hat{y}) \times (-x_1)$$

$$f(g(x)) = f'(g(x)) \times g'(x)$$